

Astera Labs – Connectivity Portfolio & Growth Within Data Centres

14 June 2024

Key Insights

- ▶ Astera has carved out a niche with its Ares product line – by developing a dominant position in the peripheral component interconnect express extension market, it has been able to capture a significant share of the GPU installations for AI training. The company will likely maintain margins for another 1-2 years, until more demand/competitors in inference emerge
- ▶ Specialist sees Astera potentially sustaining its high 45% growth for the next 1-2 years, though this is ultimately dependent on health of the training market
- ▶ The edge inferencing market for generative AI is expected to be much larger than the training market, presenting a significant opportunity for Astera Labs in the future
- ▶ Astera holds a significant share in the Compute Express Link market, but specialist sees it as tough to maintain, given competition from Broadcom and Marvel. Although this market is showing slower volume adoption and overshadowed by AI, specialist believes it will still grow as AI moves more to inference

Specialist Jeff White (JW), CTO, Edge at Dell Technologies Inc

Moderator Lucas Keh (LK), Third Bridge Sector Analyst

Agenda

- ▶ Recent IPO by Astera Labs (NASDAQ: ALAB) and market opportunity and direction recognised
- ▶ Astera Labs's connectivity offerings – Aries, Leo and Taurus
- ▶ Market share and position in data centre connectivity, highlighting growth trajectory for H2 2024 and beyond

Contents

Q: If you take a look at the growth of Astera Labs within data centres, what stands out to you?	4
Q: You mentioned that Astera has been able to establish a degree of dominance, at least from a technology standpoint, and grab some market share. Could you further break down what specific niche the company has carved out for itself? How significant of a player is it today?	5
Q: Do you think now is the correct time for Astera's IPO, which was announced in March 2024? If so, why do you see the company jumping on this opportunity now?	6
Q: What other major growth drivers would you note for Astera going forwards, outside of the data centre and edge?	7
Q: What near-term headwinds or roadblocks exist for Astera?	7
Q: On the note of the IPO, did it make more sense to break apart or are you seeing it as a missed opportunity in terms of a missed acquisition from a company such as Intel?	8
Q: For 2023, Astera reported revenue growth of about 45% YoY. How sustainable do you think this current trajectory or pace is, especially given some of the market demands and the company's positioning that you've spoken about?	8
Q: Thinking about Astera's potential growth or placement in inference, if we take a look at the product portfolio at large, could you break down the focuses for Taurus and Leo? We spoke about Aries already and the customer focus being targeted here.	9
Q: Moving forward, how do you view Astera balancing its investment or R&D focus between Taurus and Aries, for example, just given the overall slower deployment and adoption of CXL [Compute Express Link]? This is close to minimal within the data centres today.	10
Q: Do you have a range in terms of how you would determine Astera's current position or share within the CXL market?	11
Q: Would you say that Astera is able to hold a dominant or significant position, just given its technology leadership?	11
Q: You said the CXL market is being overshadowed by what's going on in AI today. From a technology or market standpoint, what catalysts do you see to ramping up that growth again?	11
Q: You mentioned a couple of times about the movement to more inference-heavy workloads. Given it seems that a lot of Astera's future growth depends on the health of the training market, what are the company's prospects in inference? How do you determine its current placement?	12
Q: Does competition change at all for Astera as the market moves towards inference from training?	12

Q: You mentioned the competitive end, as it gets a bit more sticky, and Astera obviously is up against larger companies such as Broadcom and Marvell. Although it wouldn't be the best outcome if Astera got swallowed up by this competition, do you ever view the company as a potential acquisition target in the future?	13
Q: Looking across Astera's overall revenue, we spoke a little about the product lines and segments. Where do you see the heaviest attention or concentration across the three segments? Would you say it's an even split or does one product line garner more demand?	13
Q: Where are you seeing the heaviest product traction for customers in the next 1-2 years? If we focus within Aries and Leo, where are you expecting heavier customer demand or traction? Who might these customers be?	13
Q: How much appetite are you seeing potentially from the public hyperscalers with the build outs? How much do you think Astera's future growth will depend on these types of players?	14
Q: When you take a look at the public cloud players' consumption of newer Nvidia GPUs [graphics processing units], is there a risk of a company such as Astera being left out from those newer systems?	14
Q: Can you elaborate on the upcoming Cosmos platform? What new opportunities would you say this unlocks?	15
Q: What are you going to be closely monitoring about Astera or data centre connectivity?	16

Astera Labs – Connectivity Portfolio & Growth Within Data Centres

Transcription begins at 00:00:03 of the recorded material

LK: Good afternoon. Welcome to Third Bridge Forum's Interview entitled Astera Labs – Connectivity Portfolio & Growth Within Data Centres. This is Lucas Keh and I'll be moderating today's discussion, taking a closer look into company growth developments as well as the recent IPO trajectory. Very happy to welcome Mr Jeff White. He's currently the CTO at Dell Technologies for the edge group. Jeff, thanks so much for the time, and welcome to forum.

JW: Thank you, Lucas.

LK: Before we got started with today's discussion, could you please state either I agree or I disagree to this following statement: You understand the definition of material non-public information and agree not to disclose any such information, or any other info which is confidential, during this Interview today.

JW: None of the information I provide will be confidential or violate any agreement, and everything that I share will be based on my opinion and my professional experience.

LK: For our compliance and to make things smoother, would you mind stating I agree or I disagree?

JW: I agree.

LK: Before we jump into some of these questions, we'd love to just get an introduction to yourself and your background in the space.

JW: Jeff White, work in the edge area, prior to this, I worked extensively in joint innovation research, involving AI as well as edge and other technologies. Experience in this space has been primarily in the memory fabrics and AI area, so fairly familiar with the architectures for large-scale AI model training as well as the implementation of CXL and other technologies to extend memory fabrics such as InfiniBand, etc, NVLink as well.

[00:02:23]

Q: If you take a look at the growth of Astera Labs within data centres, what stands out to you?

JW: They've been a beneficiary of the generative AI explosion that's occurred since Q4 2022 in the training area as people have scaled up infrastructures related to AI training. They've also been beneficiaries in the area of cloud and memory, the implementation of CXL, which has matured to the point where it's been commercially deployed now to solve for memory stranding in large-scale cloud infrastructures. The AI is really driving that with the interconnection of GPUs and the training and fine-tuning of large language models.

LK: Especially when it comes to that generative AI growth that we're seeing, obviously that market positioning is important. When it comes to the company's specific placement and growth when looking at LLM [large language model] training and inference, could you break this down for us?

JW: I think most of the opportunity right now is data centre-based and it's in the training area, and in the training space, I should say. There are other emerging opportunities that are also data centre-focused, as I said, to do memory optimisation and shared memory fabrics and memory extension. Those can be data centre or edge or on-site, in-premise opportunities, but those are just emerging in the memory space. I think the timing is really good for them. They've developed a dominant position. There's emerging competition that I think we'll talk about during this call, but they've been able to seize an outshare size of the market share. I think the training growth of generative models will continue for some time. The early tech adopters, the large clouds, the major technology pioneers that are building these capabilities are well on their way to either building the infrastructures or have already build the infrastructures. However, there are additional tiers of opportunity coming in the training area with sovereign AI infrastructures for governments as well as large enterprises that will start to roll these out probably more in the enterprise side, more on the fine-tuning and retrieval-augmented generation and other techniques to optimise the performance of models, both for inferencing and subsequent training optimisation. I believe that this has a tail that's going to run for a while, and then the big questions are when will inferencing kick in at scale because it will end up being ultimately much larger than the opportunity around the training environment.

[00:05:53]

Q: You mentioned that Astera has been able to establish a degree of dominance, at least from a technology standpoint, and grab some market share. Could you further break down what specific niche the company has carved out for itself? How significant of a player is it today?

JW: It's the Aries product line for them, which is really extending PCIe in GPU installations. I think that they've been a good supplier to several of the major players in that environment that need PCIe extension because the requirements in training, and I don't want to get too far in the weeds, but they need intense bandwidth, very low latency, so to be able to extend the PCIe bus and not have to do a networking conversion or some other type of conversion is very critical to keep the utilisation and GPU training up. That's a niche, I think, that they've carved out. There are some headwinds in that area in terms of people are very power-conscious, so public announcement from Nvidia at their GTC was they were going to attempt to use copper cabling or not attempt, they were going to use copper cabling and stuff in some of their larger DGX installations where they're doing rack-based DGX installations, DGX being the Nvidia AI server.

However, there's still a lot of opportunity because many other folks are going to continue to progress with PCIe because Nvidia has... their system is a very good system, it's called NVLink, but it's proprietary to them. PCIe is an open standard, and there'll still be plenty of opportunity in the PCI retimer business as well. There's other opportunity too in the cable modules and simplifying cabling in data centres, and there's growing opportunity I think we'll see outside of the cloud and folks implementing memory expansion using CXL, and they are major players there as well. There's other opportunity emerging around this that is probably not of the scale of AI training but good opportunity in their other product lines, in the Leo and additional Aries product line opportunity.

LK: I'm sure we'll touch on some of these competitive dynamics, as you mentioned, as well as the other product lines, but I wanted to ask a quick follow-up. You were saying it's pretty significant that Astera is able to extend these PCIe [peripheral component interconnect express] lanes for GPU [graphics

processing unit] installations, but when it comes to cost savings, what does that roughly look like when extending vs creating a whole new system or switching?

JW: I don't think about it in terms of cost savings, I think about it in terms of performance optimisation. Being able to extend this allows you to build bigger pods. You have GPUs and servers, typically eight GPUs per server, and then you can stack servers into pods, and then you can build the pods into clusters and super clusters. The cluster work is typically done in Ethernet or folks have even started using InfiniBand and NVLink for building clusters and super clusters, but that's the target of the Ultra Ethernet Consortium and other areas. The real extension with being able to extend PCIe several metres outside the server allows you to build a bigger pod, which is important because then you can bring more GPU horsepower directly to solve the problem and you have to worry less about check-ins and storage and synchronisation across multiple pods. It gives you a performance optimisation. The name of the game in training, the overriding objective, is to make sure the GPUs are highly utilised and are not being stalled by any check-in or anything else. When we say training, some of these training jobs can run literally for months, certainly for weeks, and so the ability not to stall them or add even a couple percentage points of inefficiency is very, very important and translates into faster time to market and much more cost-effective training and retraining.

LK: How much of that market would you say it has been able to take with Aries for that PCIe extension within the cluster type of landscape?

JW: I think it's a very large percentage. I don't know, per se. I haven't studied it from an analyst's perspective, I come more out of the technology side, but I think it's a very high percentage of that market today is based on their technology. There are other competitors out there like Rambus and Broadcom, of course, and Marvell, and they're challenging in the market, but I think they were, you could euphemistically say, the right people at the right time to leverage that capability.

[00:11:44]

Q: Do you think now is the correct time for Astera's IPO, which was announced in March 2024? If so, why do you see the company jumping on this opportunity now?

JW: I do. Simple laws of economics, when a great opportunity appears, everyone starts to try to build a better mousetrap, but the folks who built the mousetrap ahead of the great opportunity are going to benefit from it most. That's the position I think they find themselves in. They were singularly focused in this area that's grown, and they've been able to capture significant market share. What that market share percentage is, I don't know, but there are a lot of great analysts out there that I'm sure can provide information to your clients. That is a sustained market. However, competition will emerge and will start to buy into the market, and the market will change over time as it matures and it moves from a training focus into more of an inferencing approach. That's going to grow the market, so it presents future opportunity for them, but right now with their position that they've gleaned and the customers they've been able to pull in, I think that they're in a very good position to continue their growth and maintain margins for some time. As markets mature, it gets harder to maintain margins as price-based competition comes in, etc, so I think the timing is very good for them.

LK: In terms of margin preservation, do you think this will continue for the next 1-2 years? I'm trying to gauge when you're expecting that competition to have some kind of downwards pressure.

JW: In my opinion, I think it's hard to say, but I believe where that competition will begin to put margin pressure and cause them to have to compensate or maybe have less pricing flexibility would be within

the next two years and when inferencing starts to take off as these AI workloads start to move outside of the data centre and into the inferencing area. There's still going to be a demand for PCIe extension and memory fabrics and memory expansion outside of the data centre, and, ultimately, that might be a very... in my opinion, will be larger, actually, than the data centre opportunity. That will also give an opportunity for competition to position their products and grow into that space as well. Within the next two years, I'm actually fairly bullish that inferencing will start to really come into focus in 2025 and be large-scale deployment by 2026, in my opinion.

LK: Are you saying that the edge inference opportunity is significantly larger than the current data centre opportunity for Astera?

JW: No, I think just from a market perspective was what I was providing my opinion on. I think with their market position, they're well-positioned to capture that, but I also think they're going to face pretty stiff competition in this market. I do think that, ultimately, over time, it will become larger just because of the nature of the processes. You can somewhat equate model training to software development, and then you can equate inferencing to software operations or software deployment. The opportunity to inference, you train a model once, and then you may use it many, many more times than in a single instance or for a single enterprise.

[00:16:10]

Q: What other major growth drivers would you note for Astera going forwards, outside of the data centre and edge?

JW: One thing, and they have products in this area, storage is going to become very important in some of this particularly because the way the generative AI works, but also other workloads, they need a cache of information. The retrieval-augmented generation is commonly referred to, but there's a whole set of optimisations for generative AI where the models can pull information, and you can do basically prompt management and guard railing. That requires distributed storage with where you're doing the inferencing from, and they have products to do extension to storage systems and SSDs as well, so that could be an opportunity for them as well where people are starting to build distributed storage systems or more storage systems associated with the training and some of the products they've built there, their smart cable modules could play in that arena as well.

[00:17:33]

Q: What near-term headwinds or roadblocks exist for Astera?

JW: Nvidia made an announcement, at least with their DGXs, that they weren't going to be using retimers, so that's obviously a challenge, but there are a lot of folks making AI servers that will continue to focus on PCIe and not use NVLink exclusively. I think the primary challenges are not environmental, ie the growth of the market and all. I think that that's going to continue to grow, and there are other aspects of the market that will kick in over time, like the inferencing we've talked about, that will add additional size to the market. I think the primary headwind is tough competition coming from Broadcom and Marvell, Rambus to a lesser degree, that these folks are going to want to become part of that environment. I think Broadcom primarily but Marvell to a lesser degree. Broadcom has products in the space, NICs and other things that are adjacent, and they're a very focused company, a very well-run company, I think. As they look at this market space, they're going to figure out a way to compete. These

are things that they're going to have to deal with where I think they've been a bit ahead of the market, there are a lot of folks pouring money into this.

On the just general, outside of competitive headwinds, there will be a transition. The training revenue is going to start to drop, which are large-scale GPUs, which are really big deals, and it'll be more in inferencing. The deals will be smaller. They're a smaller company, they're just now IPO-ing, they've been around a while, I think they're a very capable company that seems to have good management, but they're going to have to figure out how to scale their environment beyond a few of the key players in the GPU space and work more broadly with system providers and other folks, other memory suppliers, etc. That will require them to scale their go-to-market and their customer support and all, so that'll generate additional cost for them as well. That's somewhat of the growth of any company from their size to (inaudible 20.05).

[00:20:10]

Q: On the note of the IPO, did it make more sense to break apart or are you seeing it as a missed opportunity in terms of a missed acquisition from a company such as Intel?

JW: I can't comment on the acquisition, have no knowledge of that, but I do think that the place they're in, just stepping back from AI for just a second, memory expansion, memory fabrics, load store semantic capabilities, are going to grow for a variety of reasons. AI is the head banner there, but there are other reasons that they're going to grow outside of the data centre and inside the data centre as well. They boil down to system efficiency and the ability to really evolve the compute architecture, so there's work going on in the Open Compute Project, the OCP, and others to really change the architecture of compute systems and make them a lot more modular. In the coming years, you're going to hear about modular compute systems a lot more. Again, no announcements here and nothing proprietary, but in order to build these modular compute systems, you're going to need to be able to create memory fabrics and centralised memory storage, somewhat similar to the problem that the clouds had where they had memory being stranded in servers, which was costing a lot of money, so they wanted to be able to interwork those servers so workloads could access memory on other capabilities. That problem is going to grow and grow, and it'll grow outside of the cloud, and AI will be one of the drivers for that.

That creates a real opportunity for people that know CXL, that have foundational capabilities like PCIe retimers and how to do smart cables and integration and do extension. With their knowledge base, they should be able to leverage a lot of that that goes beyond AI. I can't comment on any particular acquisition or corporate development strategy they may have, but I think they'll be pretty attractive in this space, similar to how the NICs went. NICs were developed separately, Conzando and other folks were building NICs, and NICs became foundational to management of capabilities like AWS Nitro and stuff, and then all of a sudden, a big company started to pick up NICs. I think the same thing could happen with memory fabrics. It's a possibility, and we'll see how that develops over the coming years.

[00:23:17]

Q: For 2023, Astera reported revenue growth of about 45% YoY. How sustainable do you think this current trajectory or pace is, especially given some of the market demands and the company's positioning that you've spoken about?

JW: Congratulations to them. It's a rare year you can have 45% growth in any industry in a business of any scale at all. I can't quantitatively comment on that. Quantitatively, I can comment, in my opinion, I

think they should be able to sustain high growth. Whether they can keep it at that atmospheric level or not, smarter people than I will have to answer that question. There is certainly demand, as I've mentioned. There's another wave. There are subsequent waves, beyond the early technology adopters, for AI that are going to be building out scale infrastructures. That's public knowledge and repeatedly mentioned by all the players in the AI industry, so they're going to need these kind of capabilities, and their products will be included in a lot of these AI infrastructures. It's certainly possible, in my opinion. Whether they can maintain that level or exceed it will depend upon a lot of factors beyond my control or knowledge.

LK: In terms of that sustainability and the demand in the market today, do you see that warranting the company being able to even maintain? As you say, 45% is nothing short of significant. At worst, if it were able to maintain, it would still be 45% growth. Are you seeing that kind of range as reasonable for the next 1-2 years?

JW: As I stated, I'm seeing it as possible, but it'll depend on a lot of factors. Ultimately, the training market will start to decline over time as the infrastructures are built out and as algorithmic improvements are made that lessen the need for computation. These are natural things that will happen all on their own. There are a lot of people working on improved algorithms, improved approaches to improve the efficiency of the attention transformer. It's probably top three hottest topics in AI right now is how to do that. That will slow down the infrastructure market a bit, and then, of course, it's going to mature. As it matures, then there'll be less demand. One thing I would say is there's always going to be a technology refresh need which needs to be factored in. It's a very fast-moving market and so there'll be significant capital, in my opinion, allocated to modernise and keep the capabilities growing in the training environment, so it's not going to go away by any stretch. Even as these waves of initial implementation recede, there's still going to be demand.

Beyond that, I think that the real question mark is the inferencing market. How big will that be? How much will they require Astera products and other products as they begin to distribute that? You'd earlier said edge inference. A significant component is going to be edge inferencing, I'm quite happy to say that. It's very clear that it makes a lot of sense to distribute the inferencing to where the data is being produced. However, there's also going to be a significant amount of inferencing going on in the data centre as well. That inferencing market will be, in my opinion, the determining factor in how much growth that a company like Astera can sustain. We should see that. Also, the timing on that, when does that really get to scale vs the maturity of the training market, is another key area that needs to be determined, and I don't think anyone knows that right now.

[00:27:53]

Q: Thinking about Astera's potential growth or placement in inference, if we take a look at the product portfolio at large, could you break down the focuses for Taurus and Leo? We spoke about Aries already and the customer focus being targeted here.

JW: Aries, we've talked about that, so it's used in the peripheral and components, primarily in the GPUs. Then Taurus is the Ethernet smart cable modules, so how do I scale out network and storage in fabrics and build out those capabilities, so optimising Ethernet connectivity and extend that for Ethernet, when there's a lot of work going on in Ultra Ethernet Consortium and other things at the scale-out area that is a good opportunity. Again, as we build pods and these clusters expand geographically and all, that could create opportunity in the smart cable side for Ethernet. Also, the storage, as I had mentioned, being able to build out storage stuff. Then the Leo product is the more traditional CXL and how do I extend to use memory to do memory expansion and memory centralisation? There's already business here primarily

surrounding the ability to pool memory for the clouds, but that's going to expand outside of the cloud to do memory expansion. There are some real advantages to doing memory expansion. If you look at just CXL itself, compute express link, it gets latency-bound because it's distributing the memory and so it becomes bound by latency, which can impact performance against certain workloads.

If you look at in-system DDR, it becomes latency-bound further out, so it has higher performance, obviously, because it's nearer to the computation, the CPU. However, if you interleave memory-expanded CXL as well as having core memory and you do memory interleaving where it can NUMA-type capability, place the memory in the right position, you can actually optimise both bandwidth and latency to a much greater extent than you could with just CXL only or DDR only. I think you're going to start seeing that appear because it will help memory-intensive workloads, which a lot of AI workloads are memory-intensive, being able to use these memory expanders and do memory pooling, etc. That's their Leo product line. I think they've got opportunity, actually, in Leo that's probably not realised yet that will appear. Part of it will be the AI inferencing but, also, part of that will be analytic workloads, edge workloads, that require memory intensive and high performance.

[00:31:26]

Q: Moving forward, how do you view Astera balancing its investment or R&D focus between Taurus and Aries, for example, just given the overall slower deployment and adoption of CXL [Compute Express Link]? This is close to minimal within the data centres today.

JW: It has been slower than I think anyone anticipated when CXL came out. Part of that is it's tied to PCIe, which is slower than the proprietary solutions which are largely based on InfiniBand, like NVLink, etc, and the massive, massive amount of bandwidth that very few people anticipated the GPUs would require to operate in the generative AI or the large language model frame. I think CXL has a place. The next generation of PCIe will help and gets up to 128 gigagrams per second, etc. I think it's going to be hard to manage focus. I do think that there'll be a big push, like with what's going on in the Ultra Ethernet Consortium, to really have more standardised methods of interworking the scale out of the clusters and the scale up. PCIe could play a role there, but they've got a tremendous opportunity with their Aries that's going forward. CXL has been slower, the real adoption, the commercial product roll-outs have started fairly recently, I'd say within the last 12-18 months, really. CXL could play an interesting role in this AI inferencing, and then they'll have to balance that focus and find the capital to be able to balance that focus.

LK: Do any risks stand out for Astera when it comes to that slower deployment or adoption? How do you see the company being able to adjust?

JW: There's a lot of industry momentum to try to stay away from proprietary fabrics for a variety of reasons. It's not the products aren't good, they're very good products, but if they're tied specifically to one supplier, then it's hard to interwork them and manage them, etc. There's still going to be a place for PCIe, in my opinion, and capabilities like CXL. The question is the performance level and really tied to the dynamics of how the GPU market is going to develop and how competition is going to develop there. Nvidia has done a great job of maintaining leadership there, but there's a ton of capital pouring in there into competitive play. How that develops is anyone's guess, but as it does, there may be a lot of pressure to use non-proprietary pod-level networking stuff, which a company like Astera could benefit from. CXL is a lot slower-growing, so there's more room for competition to come in there, but they, again, were early adopters and have been key in the CXL standardisation efforts and working through that. To the extent it grows, I think they have an opportunity to maintain good market share.

[00:35:15]

Q: Do you have a range in terms of how you would determine Astera's current position or share within the CXL market?

JW: I really don't. Again, I'm not an analyst. I work on the technology side, so anything I'd say would be pure speculation.

LK: In terms of product volumes, given what there is today, does Astera have a majority in that market, or roughly more than half, or is that still hard to say?

JW: I think they have a significant share of the market, I don't know if it's more or less than half. I think they have a significant share and, again, I think that that market is growing. I think there are some areas behind that are growing, but in comparison to what's happening in the AI side of the house, it's very minimal compared to that, but it is a growing market, it's probably got fairly significant growth, it's just overshadowed by AI. I think, again, my personal opinion is they're a good company, and I think that they'll be able to continue to drive that forward. There are a lot of questions about CXL and it's tie-in to PCIe, and I think within the standards community, people appreciate the challenge, and we'll see how things develop on PCIe and if it can scale to higher levels in an open environment. Again, Astera would have the wherewithal to be able to grab that business if it does.

[00:36:53]

Q: Would you say that Astera is able to hold a dominant or significant position, just given its technology leadership?

JW: It's going to be hard. They're up against folks like Broadcom and Marvell that are very large, very focused players that are well-capitalised and huge corporations. I believe their strategy of entering the public markets and doing an IPO makes sense because it'll give them access to capital to be able to more effectively compete with those folks. The generative AI growth and the explosion in generative AI has made this a very attractive market, so a lot more capital is going to pour in. They'll have to figure out a way to pace that. I don't know what their IP portfolio looks like, how much of the intellectual property they control around this stuff, but that might be a key consideration in terms of answering your question of will they be able to maintain leadership here? If they've done a good job on their IP portfolio, I think that will help tremendously.

[00:38:06]

Q: You said the CXL market is being overshadowed by what's going on in AI today. From a technology or market standpoint, what catalysts do you see to ramping up that growth again?

JW: I think as AI moves into inferencing and edge, AI is already in edge, significantly, many of the workloads that are in edge today are AI-enabled. As we move with multi-modal capabilities and a more general approach of AI into the edge, I think that that's going to drive higher performance requirements and more needs for capabilities. I think that that could drive the CXL market higher because the latency requirement will still be there, which they can address. The bandwidth requirements will be lower, so that makes it a much more PCIe-type friendly market where you may not need the proprietary InfiniBand-type base fabrics. If you think about inferencing, it still has a very tight latency requirement

because a lot of the workloads themselves are latency-sensitive. In terms of bandwidth, you're not doing a reverse patch, you're not doing the check-in, so the bandwidth requirements will be generally lower, as we deploy the server, so that makes it a much more PCIe-friendly and therefore Astera Labs-friendly environment.

[00:39:53]

Q: You mentioned a couple of times about the movement to more inference-heavy workloads. Given it seems that a lot of Astera's future growth depends on the health of the training market, what are the company's prospects in inference? How do you determine its current placement?

JW: The size of the inference market, I think there's a lot of work going on right now in that and, in my personal opinion, I believe it's going to be very large because it has to be widely deployed to gain usefulness. If you're doing a virtual agent application, for instance, you've got call centres spread around the globe to address different markets, you're going to need to deploy inferencing toward those call centres so you can do the tokenisation and embedding at the call centre and probably do guard railing and some of the rag, at least, at, say, your call centre for your virtual agent, just to give an illustrative example. Those systems will still have, as I said before, latency performance requirements and a need to centralise because the assets that are going to be doing that work are still very expensive computational assets, so you're going to want to have central capabilities networked in some sort of pod to make that happen vs building just independent systems that do that. There are other technical reasons too, in terms of data access and security and stuff that you may want to organise this into pods. I think that makes these small form factor data centres good targets for PCIe extension, etc, with capabilities in the retimer space.

[00:41:49]

Q: Does competition change at all for Astera as the market moves towards inference from training?

JW: Yes, I think it does because there's going to be pressure to combine some of these capabilities with NICs and things like that and integrate, make them more integrated vs just the GPU because, for instance, you may want to do memory pooling on the same system that you've got GPUs in that you're retiming to build a GPU pod. There may be pressure to put these kind of capabilities into a NIC or I'm just speculating, but how can you combine these capabilities in a more modular way? People like Broadcom will leap at that type of opportunity, since they're so big in the NIC space. Marvell, also, I think, plays there. There may be opportunities for other folks to try to integrate some of these capabilities into their existing product lines that may be deployed out there as well, so that creates a competitive dynamic that they'll have to manage.

LK: The company, landscape-wise, pretty much stays the same?

JW: Company can change, and I think they have competent leadership that could look at how they build into additional systems. They've already got that capability that they're building with their connectivity platform, Cosmos, and other capabilities that are looking at, "How do I make my capabilities more sticky through software, etc?" so they certainly are capable of doing other things. I think with their current product set, they've still got a pretty decent opportunity.

[00:43:47]

Q: You mentioned the competitive end, as it gets a bit more sticky, and Astera obviously is up against larger companies such as Broadcom and Marvell. Although it wouldn't be the best outcome if Astera got swallowed up by this competition, do you ever view the company as a potential acquisition target in the future?

JW: Anything can happen in a deal, and deals can always happen, so no knowledge of if that's part of their strategy or part of one of the acquiring company's strategy. As a possibility, certainly, that could happen. I think that there are just so many dynamics around a deal, it's hard to comment, but certainly is a possibility.

[00:44:52]

Q: Looking across Astera's overall revenue, we spoke a little about the product lines and segments. Where do you see the heaviest attention or concentration across the three segments? Would you say it's an even split or does one product line garner more demand?

JW: I think Aries and Leo, in my opinion, are their focus areas. They really started there. The Ethernet area is really evolving, there's a lot changing in Ethernet with the Ultra Ethernet Consortium and other areas to improve the capability of Ethernet to do super pods and super clusters and scale out. There's opportunity in Ethernet, but their background is in the load store, memory semantics and PCIe retimer space, so I think their focus is primarily in Aries and Leo. They are doing stuff in the software to manage this fleet management. Fleet management is complex, and it's done at a lot of different levels, so I don't know much about them, I've not worked with them at all on their Cosmos product, so be pretty hard to manage things from the level of the business that they're in, but certainly creating software to help manage their products and capabilities and optimise the load store semantics and things could be interesting.

LK: I'm curious in terms of how you're expecting that internal allocation to evolve. Should we still expect Aries and Leo to dominate that internal revenue driver?

JW: I don't know what their revenue breakdown is, but I think their focus for the foreseeable future will remain on Aries and Leo would be my opinion.

[00:47:08]

Q: Where are you seeing the heaviest product traction for customers in the next 1-2 years? If we focus within Aries and Leo, where are you expecting heavier customer demand or traction? Who might these customers be?

JW: I think their primary customers now are the GPU systems and memory fabrics and servers, etc. I think that'll largely remain the same. I think the mix will change a little bit as the smart memory stuff takes off, to some degree, because the memory controllers could be integrated at several different levels. I don't think their customer mix will change appreciably in the near term, unless they were to change their strategy radically.

[00:48:18]

Q: How much appetite are you seeing potentially from the public hyperscalers with the build outs? How much do you think Astera's future growth will depend on these types of players?

JW: I think that they do business with them today, the public hyperscalers, and I think that that is an opportunity for them to grow. For a variety of reasons beyond the scope of this call, I think that the hyperscalers are very focused on improving their capabilities around large-scale AI training and other capabilities, and they were already working on memory aggregation and memory pooling to optimise cost around that, that is a cost optimisation play. I think that that's an opportunity for Astera to market value-added things like Cosmos and stuff to them to see how they can help them continue to improve their performance and grab more market share in the AI training space.

LK: Again, that's dependent on training. When it comes to memory pooling, you're speaking of CXL. I'm curious if those markets slow down or come to a halt, more or less? What happens next?

JW: It's hard to say. A lot of the folks that are investing, the hyperscale clouds are not only major service providers and infrastructure-as-a-service, they're also major technology companies, so they're leading investors in AI, so I don't think those folks are going to slow down at all.

[00:50:19]

Q: When you take a look at the public cloud players' consumption of newer Nvidia GPUs [graphics processing units], is there a risk of a company such as Astera being left out from those newer systems?

JW: It's been publicly stated that folks are looking at building their own GPU capabilities. The hyperscale clouds are looking at building their own GPU and AI accelerator capabilities. Fundamentally, you still need to do PCIe retiming. I don't think folks are going to be able to move off of PCIe and Ethernet easily. These are very big companies with huge, deep pockets that can justify business cases on their own requirements, so they certainly could be working on things, they certainly are working on things that I am not aware of. Fundamentally, whether the GPU is built by Nvidia or AMD or Intel or Amazon or Microsoft or Google or whoever, they still have a need to organise them into pods and clusters due to the training process, which dictates how that will work. If they're using traditional servers, whether they built their own or purchased it from an OEM, they're still going to have a need to do PCIe retiming and extension.

LK: Does it have anything to address inference today or eventually, when this moves over? Or is it maybe too early to even consider this?

JW: The inferencing market with traditional discriminative AI is very mature today, and the hyperscale clouds have very good capabilities there with the services frameworks they've built. They have different maturity levels of how to distribute that capability, but it's easily publicly determined that they're working on things like that. I think that the demand of generative AI has really caught the entire technology industry a bit off guard, so everyone is playing catch-up, to some degree or another (audio cuts out 52.59–53.03) to the OEMs, to the clouds. I think we're all playing catch-up there right now. I think they have, in my opinion, some systemic disadvantages, when it comes to inferencing, on some of these workloads because they're centralised platforms, so they're going to have to figure out a way to extend into the edge and to on-premise, which is very hard to do. They have some disadvantages, they have some advantages, but the way that I think the edge market has been very fragmented and will

remain somewhat fragmented and hard to consolidate going into the future. I think there's a lot of opportunity in edge inferencing, I'll leave it at that.

LK: How quickly do you see Astera being able to address an opportunity with that specific type of solution when this is needed? Could we see it ramp up?

JW: Astera. Again, I think that the trajectory they're on addresses a lot of these requirements because it's a foundational requirement, really. Inferencing is very similar to training, without a backpropagation pass, so their capabilities will be pretty useful in inferencing. In my opinion, they may need different scale levels. They may need smaller packaging or other type of capabilities to address the distribution of these clusters to get a better price fit for a smaller-scale pod, but packaging is not a hard thing to change, you just need to have the demand to build a business case. I think from a technical perspective, most of their stuff is foundational and will be applicable into inferencing with fairly minimal changes, in my opinion.

[00:55:21]

Q: Can you elaborate on the upcoming Cosmos platform? What new opportunities would you say this unlocks?

JW: The way the shared memory systems work and, to some degree, the way the GPUs work is they're mapping data into memory. There's software running above some sort of operating system that's accessing that memory to pull in tensors and run matrix multiplication in training or inferencing in AI, but there are tons of other applications in analytics and many other areas. The way that that memory is mapped and interleaved has a lot to do with the performance, and that's done algorithmically in the memory controllers and in the CPUs themselves, but they only have a certain view, a certain perspective. The CPU can only see what it's directly addressing through its cache and its local memory. The memory controllers don't understand what's all going on inside the CPUs with speculative instruction, execution and other things. Having something like Cosmos that can optimise that could be powerful because it could improve performance in limited resource conditions and improve latency and resource utilisation. The closest example I could say is NUMA, non-uniform memory access, where you have dual-socket servers that both have access to memory but some memory is much lower latency than other memory in the system where it may have to go through an interconnect or some sort of chip-to-chip connect, etc.

You want to map the software and data that's being accessed by that CPU nearest to it that you can so you can meet the latency requirements. Cosmos, I believe, I haven't worked with it directly, but I believe that it has some of the ability to do that, and that could be useful. I think that would be something that if it's doing that, would be very useful. Also, understanding where you've got performance problems and predictive failures and other things like that are going to be very important. These systems are super expensive, GPUs have been publicly stated... could be USD 30,000–50,000 a GPU. You want to have high utilisation there, but they also have very, very high integration, hundreds of billions of transistors, tens of billions to hundreds of billions of transistors. Being able to predict failures ahead of time and all is also useful. Again, haven't seen Cosmos directly, but I think that it's targeted at somewhat of these areas to help optimise performance, and that could be very useful and tie to them. However, it probably would be proprietary to Astera, so I don't know if they're going to make that work across other people's platforms too.

LK: Do you see that being a significant revenue driver for 2024 or more for 2025?

JW: Don't know, wish I could tell you.

LK: How about overall adoption?

JW: Overall adoption, I think in terms of adoption curve, again, I really can't comment, I don't know the maturity of it, I haven't worked directly with the platform. I've heard of it, I've talked to folks who've mentioned it, but I don't know where they are in their product maturity or their adoption at all.

LK: Who could be the most likely users or customers for this?

JW: I think that it could be used at the system level by the clouds to help optimise their performance system, so I would probably rank them near the top of the list. Other folks could incorporate that into their fabrics and pod designs as well, either at the OEM system level for folks that are trying to build bespoke infrastructures or other private cloud folks that are building not public hyperscale but private AI clouds, and then, lastly, OEMs could potentially use this, too, as part of a reference design.

[01:00:29]

Q: What are you going to be closely monitoring about Astera or data centre connectivity?

JW: What will I be monitoring closely? I think Astera is a company that has focused, they've had focus in this area, they started at the right time, and I wish them all the luck in the world.

LK: Jeff, appreciate you joining us today for our discussion. As we approach our hour's end, I think this would be a great place to wrap up. I'll close today, and thank you, again, for your insights. Hope you enjoyed it as well.

JW: Thank you, Lucas. I did.

LK: Thanks and take care.

Transcription ends at 01:01:15 of the recorded material

If you'd like to speak to Jeff White in a private call or meeting, please let your relationship manager know.

Disclaimer

The information, material and content contained in this transcript ("Content") is for information purposes only and does not constitute advice or an offer or inducement of any type or a trade recommendation and should not form the basis of any investment decision. This transcript has been edited by Third Bridge and may differ from the audio recording of the Interview. Third Bridge Group Limited and its affiliates (together "Third Bridge") make no representation and accept no liability for the Content or for any errors, omissions or inaccuracies in respect of it. The views of the specialist expressed in the Content are those of the specialist and they are not endorsed by, nor do they represent the opinion of, Third Bridge. Third Bridge reserves all copyright, intellectual and other property rights in the Content. Any modification, reformatting, copying, displaying, distributing, transmitting, publishing, licensing, creating derivative works from, transferring or selling any Content is strictly prohibited.